

PRAKHAR SINGH

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Education

BTech, Mathematics and Computing	Delhi Technological University (Formerly DCE)	2017 - 2021
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Experience

Software Developer	CreateBytes	Mar 2023 - Now (1 year, 8 months)
AI Fitness Tracking System	- Developed an web-based AI fitness tracking system using MediaPipe for real-time pose estimation, providing instant feedback on exercise form and technique. - Designed and implemented an innovative database schema for storing exercise recognition logic, enabling dynamic registration and management of new exercises without code modifications. - Created state management systems for exercise tracking by analyzing skeletal key points and joint angles in real-time.	
Naval Logistics Application (ASI)	Developed a GUI application for deployment on ships and submarines using C++ and Qt, enabling offline report generation and synchronization with ship logistics systems upon internet access.	
3D Body Analysis Tools (Proof Of Concept Apps)	- Created a PyQt GUI application that processes 3D human body scans to extract anthropometric parameters. - Developed a real-time SMPL mesh deformation tool with a PyQt-based GUI interface.	
Computational Geometry Project (Under NDA)	- Developed and implemented advanced C++ functions for deforming 3D point clouds. - Optimized spatial queries using C++, improving performance for large geometric data. - Implemented Delaunay Triangulations and Voronoi Diagrams in C++ for Computational Geometry and Computer Graphics applications. - Utilized Visual Studio, Visualization Toolkit (VTK), Qt/QML, C++17, CMake, Boost, and Eigen libraries.	

Software Consultant	Radicle Inc	Jun 2022 - Mar 2023 (10 months)
- Improved functionality of a dynamic web application by implementing enhancements using ASP.NET (MVC) with Entity Framework/C# backend and utilizing LINQ. - Streamlined database operations by optimizing database calls, eliminating code redundancy. - Created visually appealing and responsive UIs using HTML, CSS, Bootstrap, JavaScript, and jQuery. - Implemented efficient functions and queries for evaluating financial instruments such as revenue, budget, and P&L, significantly decreasing mathematical rounding errors in calculations, ensuring accuracy and precision.		

Software Developer Intern	Xtreme Technologies	Mar 2022 - Jun 2022 (4 months)
- Developed a GUI application using the Qt framework (C++) from the ground up. - Developed mathematical functions for visualizing color gradients and isolines from raw gridded data. - Designed an encoder/compressor solution, significantly reducing gridded data size and enhancing storage efficiency through binary files.		

Projects

CBIR System Using Radial Colour Histograms	- Developed a CBIR (Content Based Image Retrieval) system. - Proposed a photo-radial colour histogram about the centroid of the grey-scale image for classification. - Worked with Python (numpy, cv2, matplotlib), Jupyter Notebook
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Skills

Programming Languages	C/C++, Python, C#, JavaScript
Libraries	Boost, Eigen, tinyxml, Vtk/C++, PyVtk, o3d, Mediapipe, opencv
Frameworks	Qt/C++, PyQt, Visual Studio, CMake
Academic Coursework	Linear Algebra, Computer Graphics, Computational Geometry